# YIFAN (STEVE) LYU

NLP & Trustworthy LLMs | PhD Applicant

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Google Scholar | GitHub

**RESEARCH INTERESTS** Social NLP · LLM Behavioral Evaluation · Structured NLP

I study how language models use linguistic and social context, combining controlled behavioral evaluation with structured system design to diagnose failure modes and build more reliable systems.

## EDUCATION

**Dalian University of Technology** — Dalian, China

M.S. in Software Engineering, School of Software — Sep 2024 - expected Jun 2027

GPA: 3.63/4.00 | Average: 85.76/100

U.S. News 2026: #34 in Artificial Intelligence

Supervisor: Xiujuan Xu, Associate Professor

**South-Central Minzu University** — Wuhan, China

B.Eng. in Software Engineering — Sep 2020 - Jun 2024

GPA: 3.36/5.00 | Average: 84.15/100

Bachelor of Management in Financial Management

Thesis supervisor: Meng Shi, Associate Professor

## PUBLICATIONS AND PREPRINTS

### Peer-Reviewed Publications

1. Xinran Li, Xinze Che, **Yifan Lyu**, Zhiqi Huang, and Xiujuan Xu. "TCDA: Thread-Constrained Discourse-Aware Modeling for Conversational Sentiment Quadruple Analysis." *IJCAI-ECAI 2026 Main Track*, accepted.
   Models conversational sentiment quadruples by combining discourse-aware positional encoding with thread-constrained dialogue structure.
2. Xiujuan Xu, **Yifan Lyu**, Guanghui Li, and Zhehuan Zhao. "Time-Varying Propagation Graph Construction for Semantic Model Dissemination." *IEEE SmartIoT 2026*, accepted.
3. Xiujuan Xu, Guanghui Li, **Yifan Lyu**, and Zhehuan Zhao. "Nonlinear Propagation Mechanisms of Semantic Models in Complex Topologies." *IEEE SmartIoT 2026*, accepted.

### Preprints

4. **Yifan Lyu**, Dianqing Lin, Xinran Li, Jiaqi Qiao, and Xiujuan Xu. "SPAR-Hate: Auditor-Guided Multi-Perspective Role Reasoning for Bilingual Hate Speech Parsing." arXiv preprint, 2026.
   Presents an auditor-guided multi-perspective role-reasoning framework for bilingual hate-speech parsing with evidence-constrained arbitration and sample-level reassembly.
5. **Yifan Lyu**, Xinran Li, Jiaqi Qiao, and Xiujuan Xu. "Signal or Spurious Cue? A Randomized Audit of Survey-Country Metadata in LLM Social Inference." arXiv preprint, 2026.
   Tests whether verified survey-country metadata improves LLM forecasts and whether randomly assigned country labels steer predictions.
6. Jiaqi Qiao, Xinran Li, **Yifan Lyu**, Xiujuan Xu, and Liu Yu. "Expert-Guided Multimodal Fusion for Unified Emotion and Sentiment Analysis." arXiv preprint, 2025.
7. Xinran Li, Xiaowei Zhao, Yubo Zhu, Zhiheng Zhang, Zhiqi Huang, Hongkun Song, Jinglu Hu, Xinze Che, **Yifan Lyu**, Yong Zhou, and Xiujuan Xu. "Extensible Multi-Granularity Fusion Network and Transferable Curriculum Learning for Aspect-based Sentiment Analysis." arXiv preprint, 2024.

## RESEARCH EXPERIENCE

**SPAR-Hate: Auditor-Guided Multi-Perspective Role Reasoning for Bilingual Hate Speech Parsing**

Dalian University of Technology | Research Lead — May 2026 - Jul 2026

- Independently conceived and led the development of an auditor-guided multi-perspective role-reasoning framework for bilingual structured hate-speech parsing, combining local focus-unit decomposition, evidence-constrained arbitration, and sample-level reassembly.

- Designed the experimental protocol and bilingual data preparation; conducted baseline comparisons, component ablations, and bounded-arbitration diagnostics to assess the contribution of each reasoning component.
- With the same Qwen3-14B backbone, SPAR raised the reported average across main-track parsing metrics from 29.83 to 32.82 on ZH and from 29.59 to 37.51 on EN, relative to a zero-shot baseline.

**Signal or Spurious Cue? A Randomized Audit of Survey-Country Metadata in LLM Social Inference**

Dalian University of Technology | Research Lead Jun 2026 - Jul 2026

- Designed and executed a provenance-controlled behavioral audit separating verified metadata utility from directional uptake of randomly assigned country labels in LLM individual-response forecasts.
- Built an auditable inference pipeline spanning five LLMs, six countries, seven targets, and four metadata conditions.
- Constructed PROV-FORECAST with 14,400 paired probability distributions. On the separate 72-record panel, verified-country metadata reduced held-out Brier loss by 0.040 (95% CI [0.024, 0.056]); the estimated reduction in country-directed shifts after disclosing randomized-label provenance was 0.0003 (95% CI [-0.0157, 0.0166]).

**TCDA: Thread-Constrained Discourse-Aware Modeling for Conversational Sentiment Quadruple Analysis**

Dalian University of Technology | Research Collaborator Jan 2026 - Apr 2026

- Proposed and designed Discourse-Aware Rotary Position Embedding (D-RoPE), introducing topology-adaptive rotary encoding at token and utterance levels for conversational sentiment quadruple analysis.
- D-RoPE improved the MVQPN baseline from 35.68 to 37.52 Micro-F1 on ZH and from 35.62 to 36.42 on EN; the full TCDA model achieved 44.35 Micro-F1 on ZH and 39.69 on EN.

**Expert-Guided Multimodal Fusion for Unified Emotion and Sentiment Analysis**

Dalian University of Technology | Research Collaborator Oct 2025 - Jan 2026

- Contributed to prompt wrapping and pseudo-token injection; conducted LoRA fine-tuning experiments on MELD and MOSEI, plus four-backbone comparisons and modality ablations across MELD, CHERMA, MOSEI, and SIMS-V2.

**Cross-Modal Emotion Understanding: A Transformer-GAT Approach for Dialogue Emotion Recognition**

Dalian University of Technology | Research Collaborator Oct 2024 - Dec 2024

- Contributed EmoBERTa-based text feature extraction and modality ablations on IEMOCAP and MELD.

## HONORS & AWARDS

- **First Prize**, National English Competition for College Students (NECCS), Non-English Major Category, 2022.
- **Academic Scholarship**, South-Central Minzu University, 2023 and 2024.
- **Honorable Mention**, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM), 2022.
- **College-level Inspirational Scholarship**, South-Central Minzu University, 2021.

## TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch · LLM Engineering: vLLM, PEFT (LoRA/QLoRA), SFT, RLHF/GRPO, prompt design and debugging, API automation · Tools: Git/GitHub

## RESEARCH METHODS & ANALYSIS

Experimental Design & Evaluation: hypothesis-driven experiments, controlled evaluation, behavioral auditing · Modeling & Analysis: dataset construction/alignment, structured prediction, ablation/error analysis · Statistical Inference: bootstrap CIs, permutation tests

## TEACHING EXPERIENCE

**Teaching Assistant**, Artificial Intelligence and Compiler Technology | Sep 2024 - Jan 2025

Dalian University of Technology

- Graded coursework and guided hands-on compiler laboratories.

## LANGUAGES

**Mandarin Chinese:** Native; PSC 93.1/100 (Grade 1-B).

**English:** IELTS Academic 8.0 (Listening 8.5; Reading 8.5; Writing 6.5; Speaking 7.5), January 2026.

**Additional Languages:** Korean (elementary); Spanish (beginner).